shugaba

FOR ERIK, DANIEL AND KHALID

Build plan & timeline

What needs to be done, who does it, and by when. 14 days from today to submission.

VERSION	SUBMIT BY	DEVPOST CLOSES
1.0 · Thu 29 Oct 2026	Wed 11 Nov, 6 pm Lagos	Thu 12 Nov, 9 pm Lagos

PayPal AI Hackathon: what matters

Re-checked on 29 Oct 2026 against the Devpost page, rules and dates page (paypalaihackathon.devpost.com).

12 Nov

Submissions close 12:00 pm PST = 9:00 pm Lagos. We aim for Wed 11 Nov, 6 pm.

1–15 Dec

Judging period. The demo URL, sandbox and licences must work the whole time.

21 Dec

Winners announced (on or around, 2 pm PST).

How we are judged

Stage one is pass/fail: it must fit the theme and **meaningfully use PayPal and AI**. Stage two scores five criteria, **equally weighted**. Ties go to the higher score on the first criterion, then the next, so Technological Implementation is the tiebreaker.

Criterion	How Shugaba earns it
Technological Implementation	Real sandbox money loop (Vault, Orders, Payouts, Invoicing, webhooks), hash-chained ledger, property tests, agents behind hard validators
Design	Minimal white UI, Pebble as a character, live calendar and control tower, every state designed
Potential Impact	Savings circles are used by millions across West Africa and the diaspora; the guarantee and score fix real trust problems. Cite every number
Innovation / Idea	Protection Pool, agents negotiating turn swaps, a public ledger anyone can verify
Presentation	Sub-3-minute video, judge panel that runs the whole story, README that stands alone

What we must submit

- [ ] Public GitHub repo with all code, assets and run instructions, and an open source licence (MIT) shown in the repo's About section
- [ ] Text description of features, plus which AI tools and models we used and how
- [ ] Working demo: hosted URL with test logins (our `/judge` panel) free and open to judges through 15 Dec
- [ ] Demo video under 3 minutes, public on YouTube, showing the real app, no copyrighted music or third-party trademarks without permission
- [ ] Devpost form: sponsor tools ticked (AG Grid, Bryntum, Render), all three of us added as teammates

Prizes we can win

One Grand Prize **or** one Honorable Mention, **plus** one Sponsor Prize.

Prize	Amount	Our angle
1st / 2nd / 3rd overall	$12k / $8k / $5k	The main target
Honorable Mentions	$5k each	Best Use of PayPal + AI, Best Use of Agentic Commerce (Turn Market), Most Impactful, Best Demo Delivery

Prize	Amount	Our angle
Best Use of AG Grid	$5k / $2k / 3 × $1k	Control tower in AG Studio. Biggest sponsor prize
Best Use of Bryntum	3 × $1k	Payout calendar members actually use
Best Use of Render	$1k / $750 / $500 credits	Hosted demo on Render

02 · TEAM

Who owns what

Erik builds the spine. Daniel and Khalid each own a lane that can be built and tested on its own, against shared types and mocks, so nobody waits on anybody. Anyone can pick up anything; these are owners, not walls.

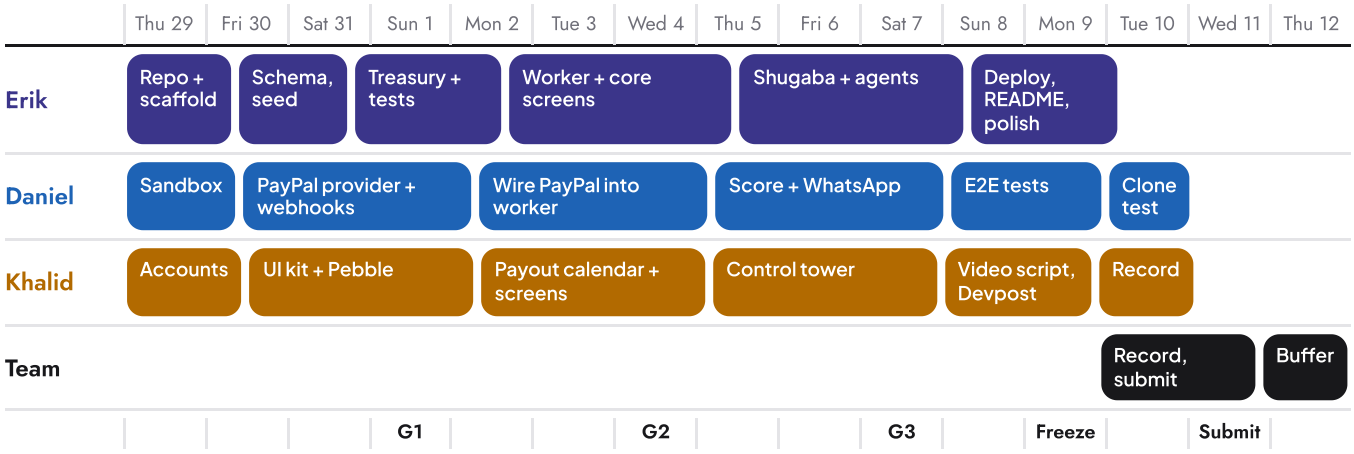
	Owns	Works against
Erik	Repo, data model, treasury rules, worker, core screens, Shugaba and the agents, deploy	The build spec, milestone by milestone
Daniel	PayPal lane: sandbox setup, `PayPalProvider`, webhooks, payouts, invoices, reconciliation. Then Circle Score + smart order, WhatsApp, Playwright E2E	`PaymentProvider` interface + `MockProvider` contract tests; `src/lib/score` is pure functions
Khalid	UI lane: design tokens, UI kit, Pebble component, avatars and levels, payout calendar (Bryntum), control tower (AG Studio). Then video, Devpost, screenshots	Fixture data on a `/dev/kit` page; calendar and widgets take props, so they work before the backend does

How we work

- **One repo, small PRs into `main`.** Branch per task, CI green before merge, Conventional Commits. `main` must always run with zero keys.
- **Erik pushes `src/lib/protocol` types first** (Fri 30 Oct, midday). Everyone builds on those shapes.
- **One short check-in a day** (15 min, or a message): done, next, blocked.
- **Secrets** go in Render env and a private password manager, never in git or chat.

14 days at a glance

Four gates. Each one is a hard check: if it fails, we cut scope (section 05) before moving on.



Gate	When	Pass means
G1 · Foundations	Sun 1 Nov, night	Treasury tests pass (≥ 1,000 property runs). A script charges a vaulted sandbox buyer, pays out and sends an invoice. UI kit page shows Pebble, avatars and levels
G2 · Money loop	Wed 4 Nov, night	Golden path steps 1–7 click through in the browser on the Render URL (mock), and in sandbox locally. UptimeRobot is pinging
G3 · MVP	Sat 7 Nov, night	All 11 golden-path steps run on Render in sandbox, by someone who didn't build them, twice in a row
Freeze	Mon 9 Nov, 11:59 pm	No new features. Only P0/P1 fixes, recording and submission

This starts from an empty repo today. The old playbook had Gate 1 (PayPal loop in sandbox) on 29 Oct; if any of that already exists, pull the dates forward and use the gain as buffer.

Tasks

Day 0 · Set up

Thu 29 Oct

Erik Create the GitHub repo (public, MIT). Commit the spec as `docs/BUILD_SPEC.md` and the design folder as `docs/design/`. M1: Next.js app, Supabase CLI config, worker entry, CI, `.env.example`

Daniel PayPal sandbox app: platform business account + 6 member buyers (Amina, Musa, Ngozi, Tunde, Ifeoma, Sani) + 2 spares for judges. Check Vault is enabled; this decides `COLLECTION_MODE`

Khalid Register all three on Devpost; join the PayPal Discord. Create the Supabase free project, Render service, and put the Gemini key in Render env. Get Bryntum trial access and the AG Studio boilerplate + licence key. Meta developer app with the WhatsApp test number and our phones

Days 1–3 · Foundations

Fri 30 Oct – Sun 1 Nov

Erik M2: protocol schemas (push first), Drizzle schema, RLS, idempotent seed. M3: treasury pure functions (fees, settlement, pool cover, state machines, ledger hash chain, trade validation) with unit + property tests

Daniel M4: `PayPalProvider`: OAuth cache, vault setup/confirm, vaulted charge, orders, payouts, invoices, transactions. Webhook verify + dedup. Contract tests shared with `MockProvider`. Leave `// VERIFY:` links on every call

Khalid Tokens + Tailwind preset; fonts; UI kit (buttons, hairline lists, sheets, ask bar, amount text). Pebble component with 7 states + breathing/blink. Avatars with 10 badges, level perch. Fallback CSS-grid payout calendar on fixture data

G1 Foundations pass (gate table in section 03).

Days 4–6 · Money loop

Mon 2 – Wed 4 Nov

Erik M5: pg-boss worker (clock, collect, grace, settle, payout, recover), REST route handlers, Realtime broadcasts. M6: auth + demo picker, home, create, join, circle home (ring), ledger, verifier, judge panel. First Render deploy + `/api/health`

Daniel Plug `PayPalProvider` into the worker; `/api/webhooks/paypal`; run golden steps 1–7 against sandbox. Nightly reconciliation job. Set up UptimeRobot

Khalid M7: Bryntum Scheduler behind the `PayoutCalendar` adapter, themed, live updates, drag-to-swap request. Feed, score and notifications screens. Loading, empty, error, offline, “PayPal is down” states

G2 Money loop passes.

Days 7–9 · Differentiators

Thu 5 – Sat 7 Nov

Erik M8: `llm.ts` (Gemini + mock), Shugaba chat in en/pcm/ha with confirm cards, recovery agent, member agents + Turn Market, narration, evals incl. the 3 prompt-injection cases. `pnpm record:golden` so judges’ runs use zero quota

Daniel M9: Circle Score, levels, smart order (pure + tests; Amina reaches Pillar). WhatsApp channel: mock first, then the test number with the 5 utility templates

Khalid M10: control tower in AG Studio (pool coverage, risk heatmap, cash-flow waterfall, recovery tracker, fraud flags, agent traces), AG Grid Community fallback. Operator-only `/ops`

G3 MVP on Render, whole golden path, twice.

Days 10–11 · Harden and freeze

Sun 8 – Mon 9 Nov

Erik	M11: `render.yaml`, README (section 17 outline), security checklist (spec 4.2). M12: Hausa + Pidgin on core screens, 390px pass. Analyst agent only if everything else is green
Daniel	Playwright golden path in mock mode in CI. Webhook replay and bad-signature tests. Check sandbox accounts and webhooks will stay active to 15 Dec
Khalid	Video script and shot list (target 2:50). README screenshots + calendar GIF. Devpost draft: tagline, story, built-with, 4+ gallery images, testing instructions

Freeze Mon 9 Nov, 11:59 pm.

Days 12–14 · Ship

Tue 10 – Thu 12 Nov

Khalid	Tue: record the video with Erik driving the app. One clear voice, captions, no copyrighted music
Daniel	Tue: fresh-clone setup test from the README, on a machine that didn't build it. Must run in under 15 minutes
Erik	Tue: fix P0/P1 only. Wed: final README, tag `v1.0-submission`
Team	Wed 11 Nov: video public on YouTube, submit on Devpost by 6 pm Lagos. Thu 12: buffer only, Devpost closes 9 pm Lagos

05 · IF WE FALL BEHIND

Cut list, in order

Cut from the top. Never cut the bottom block: that is what the judges must see.

Cut first

1. Analyst agent in the control tower
2. WhatsApp on the real test number (keep mock preview)
3. Hausa and Pidgin UI strings (keep Shugaba speaking them)
4. 3D Pebble (keep the flat one)
5. AG Studio → AG Grid Community fallback
6. Bryntum → fallback grid calendar
7. Sandbox Vault → buyer-approved order per round

Never cut

- Real PayPal sandbox money in and out
- Pool covers a missed payment, payout stays full
- Shugaba chat, with a refusal
- Turn swap through member agents
- Ledger that verifies in the browser
- Judge panel that runs it all alone

Bug triage after freeze. P0: breaks the golden path or moves wrong money, fix now. P1: visible in the video, fix before recording. P2: list under known issues in the README.

06 · RISKS

What could hurt us

Risk	What we do
Vault not enabled on our sandbox app	Daniel checks on Day 0. Fallback: buyer-approved order per round; note it in `docs/DECISIONS.md`
Gemini free quota is tiny (~20 Flash calls a day)	Flash-Lite for fast tasks, 10-minute cache, record-and-replay for the golden path. Only free chat goes live
Render free sleeps, Supabase pauses	UptimeRobot every 5 min + GitHub Actions backup ping, kept on until 21 Dec. Ask Render about sponsor credits for a paid instance
Bryntum / AG Studio trials end ~12 Dec, judging runs to 15 Dec	Ask both sponsors for a licence through judging now. Fallbacks ship anyway
Hausa copy is wrong	Find a native speaker by Thu 5 Nov to review everything marked `NEEDS_NATIVE_REVIEW`
Uncited numbers in the pitch	Khalid sources every statistic (World Bank Findex, central bank reports) or we drop it
Demo login picker leaks to a real deployment	Boot and build fail when `DEMO_MODE` meets live PayPal or an unlisted host (spec 4.2)

07 · OPEN ITEMS

Decide this week

- [ ] Who is our Devpost representative (presses submit)?
- [ ] Collection mode: Vault or order-per-round (after Daniel's Day 0 check)
- [ ] Who voices the video?
- [ ] Native Hausa reviewer: who and by when
- [ ] Sponsor licences past trial, and Render credits
- [ ] Confirm PayPal fee rates per country so the fee preview is right

08 · AFTER WE SUBMIT

Keep it alive until 21 Dec

- Don't push to the judged deployment after submitting. Work, if any, goes on a branch.
- Keep UptimeRobot, the Render service, the Supabase project, the PayPal sandbox app and the WhatsApp token running through judging (1–15 Dec).
- Run the golden path from `/judge` once a week and after any sponsor licence change.

09 · REFERENCES

Where things are

What	Where
Build spec	`adashi/SHUGABA_BUILD_SPEC.md` in the project folder; copy into the repo as `docs/BUILD_SPEC.md`
Tech stack + design system	`adashi/docs/shugaba-tech-stack-and-design-system.pdf`
Screen designs	`adashi/design/` in the project folder; copy into the repo as `docs/design/`
Hackathon	Devpost overview · Rules · Dates
PayPal docs	developer.paypal.com (check before every call)